1 MATRITSALAR. MATRITSALAR USTIDA AMALLAR.

1.1. Quyidagi matritsalarning $2A+3B$ chiziqli kombinatsiyani toping, bu yerda

$$A=\begin{pmatrix}1 & 2 & 3\\ 0 & 1 & -1\end{pmatrix},\ B=\begin{pmatrix}-2 & 3 & 0\\ 2 & 1 & 1\end{pmatrix}.$$

Yechish.

$$2A+3B=2\cdot\begin{pmatrix}1 & 2 & 3\\ 0 & 1 & -1\end{pmatrix}+3\cdot\begin{pmatrix}-2 & 3 & 0\\ 2 & 1 & 1\end{pmatrix}=\begin{pmatrix}2 & 4 & 6\\ 0 & 2 & -2\end{pmatrix}+$$

$$+\begin{pmatrix}-6 & 9 & 0\\ 6 & 3 & 3\end{pmatrix}=\begin{pmatrix}2-6 & 4+9 & 6+0\\ 0+6 & 2+3 & -2+3\end{pmatrix}=\begin{pmatrix}-4 & 13 & 6\\ 6 & 5 & 1\end{pmatrix}.$$

?

Berilgan matritsalarning chiziqli kombinatsiyasini toping.

1.2. $A-\lambda E,\ A=\begin{pmatrix}2 & -1 & 2\\ 5 & -3 & 3\\ -1 & 0 & -2\end{pmatrix}.$

1.3. $4A-5B,\ A=\begin{pmatrix}2 & -1 & 0\\ 3 & 4 & -2\\ -3 & 1 & 5\end{pmatrix},\ B=\begin{pmatrix}3 & 1 & 2\\ -2 & 1 & 3\\ 0 & 2 & -4\end{pmatrix}.$

1.4. $3A+4B,\ A=\begin{pmatrix}7 & -2 & 3 & -4\\ 0 & 2 & 1 & -1\\ -5 & 3 & 2 & 0\end{pmatrix},\ B=\begin{pmatrix}2 & -1 & -3 & 1\\ 7 & -1 & 0 & 4\\ 8 & -2 & 1 & 5\end{pmatrix}.$

1.5. $2A-B,\ A=\begin{pmatrix}1 & 5\\ 2 & -4\end{pmatrix},\ B=\begin{pmatrix}3 & 2\\ 4 & 1\end{pmatrix}.$

1.6. $3A-2B,\ A=\begin{pmatrix}1 & -1 & -3\\ 2 & 1 & 5\end{pmatrix},B=\begin{pmatrix}0 & 3 & 2\\ -1 & 4 & 1\end{pmatrix}.$

1.7. $2A+5B,\ A=\begin{pmatrix}3 & 5\\ 4 & 1\end{pmatrix},\ B=\begin{pmatrix}2 & 3\\ 1 & -2\end{pmatrix}.$

1.8. $A=\begin{pmatrix}1 & 2 & 3\\ 1 & 0 & -1\end{pmatrix}$ va $B=\begin{pmatrix}3 & 4 & 5\\ 6 & 0 & -2\\ 7 & 1 & 8\end{pmatrix}$ matritsalar berilgan. AB va BA matritsalar ko‘paytmasi (agar mumkin bo‘lsa)ni toping.

Yechish. $AB=\begin{pmatrix}1 & 2 & 3\\ 1 & 0 & -1\end{pmatrix}\cdot\begin{pmatrix}3 & 4 & 5\\ 6 & 0 & -2\\ 7 & 1 & 8\end{pmatrix}=$

$$=\begin{pmatrix}1\cdot3+2\cdot6+3\cdot7 & 1\cdot4+2\cdot0+3\cdot1 & 1\cdot5+2\cdot(-2)+3\cdot8\\ 1\cdot3+0\cdot6+(-1)\cdot7 & 1\cdot4+0\cdot0+(-1)\cdot1 & 1\cdot5+0\cdot(-2)+(-1)\cdot8\end{pmatrix}=$$

$$=\begin{pmatrix}36 & 7 & 25\\ -4 & 3 & -3\end{pmatrix}.$$

BA ko‘paytma mavjud emas, B matritsaning ustunlari soni A matritsaning satrlari soniga mos kelmaydi

? AB va BA matritsalar ko‘paytmasi (agar ular mavjud bo‘lsa)ni toping.

1.9. $A=\begin{pmatrix}3 & -2\\ 5 & -4\end{pmatrix},\ B=\begin{pmatrix}3 & 4\\ 2 & 5\end{pmatrix}.$

1.10. $A=\begin{pmatrix}4 & 0 & -2 & 3 & 1\end{pmatrix},\ B=\begin{pmatrix}3\\ 1\\ -1\\ 5\\ 2\end{pmatrix}.$

1.11. $A=\begin{pmatrix}1 & 2\\ 3 & 6\end{pmatrix},\ B=\begin{pmatrix}2 & 6\\ -1 & -3\end{pmatrix}.$

1.12. $A=\begin{pmatrix}1 & 2 & 3\end{pmatrix},\ B=\begin{pmatrix}4 & -3\\ 1 & 2\\ 0 & 2\end{pmatrix}.$

1.13. $A=\begin{pmatrix}2 & 1 & -1\\ 0 & 1 & 0\\ 0 & 0 & -1\end{pmatrix},\ B=\begin{pmatrix}1 & -1\\ 0 & 1\\ 1 & 0\end{pmatrix}.$

1.14. $A=\begin{pmatrix}1 & -1 & 2\\ 2 & 3 & 4\\ -4 & 5 & 1\end{pmatrix},\ B=\begin{pmatrix}3 & 4 & 1\\ 0 & 2 & 5\\ 1 & -1 & 4\end{pmatrix}.$

1.15. $A=\begin{pmatrix}1 & -1 & 3\\ 2 & 1 & 5\end{pmatrix},\ B=\begin{pmatrix}1\\ 2\\ 3\end{pmatrix}.$

1.16. $A=\begin{pmatrix}1 & 3 & -1\\ 2 & 1 & 2\\ 0 & 1 & 0\end{pmatrix},\ B=\begin{pmatrix}1 & 1\\ 2 & 3\\ 1 & 0\end{pmatrix}.$

1.17. $A=\begin{pmatrix}2 & 0 & 1\\ -2 & 3 & 2\\ 4 & -1 & 5\end{pmatrix},\ B=\begin{pmatrix}-3 & 1 & 0\\ 0 & 2 & 1\\ 0 & -1 & 3\end{pmatrix}.$

1.18. $A=\begin{pmatrix}1 & -3 & 0\\ 2 & 5 & 1\end{pmatrix},\ B=\begin{pmatrix}0 & -1 & 3\\ 3 & 5 & 2\\ 4 & -2 & 1\end{pmatrix}.$

1.19. $A=\begin{pmatrix}1 & -3 & 2\\ 3 & -4 & 1\\ 2 & -5 & 3\end{pmatrix},\ B=\begin{pmatrix}2 & 5 & 6\\ 1 & 2 & 5\\ 1 & 3 & 2\end{pmatrix}.$

1.20. $A=\begin{pmatrix}2 & -1 & 3 & -4\\ 3 & -2 & 4 & -3\\ 5 & -3 & -2 & 1\\ 3 & -3 & -1 & 2\end{pmatrix},\ B=\begin{pmatrix}7 & 8 & 6 & 9\\ 5 & 7 & 4 & 5\\ 3 & 4 & 5 & 6\\ 2 & 1 & 1 & 2\end{pmatrix}.$

1.21. $A=\begin{pmatrix}5 & 7 & -3 & -4\\ 7 & 6 & -4 & -5\\ 6 & 4 & -3 & -2\\ 8 & 5 & -6 & -1\end{pmatrix}$, $B=\begin{pmatrix}1 & 2 & 3 & 4\\ 2 & 3 & 4 & 5\\ 1 & 3 & 5 & 7\\ 2 & 4 & 6 & 8\end{pmatrix}$.

1.22. $\begin{pmatrix}1 & 1\\ 0 & -1\end{pmatrix}^9$ matritsani toping.

Yechish. $A=\begin{pmatrix}1 & 1\\ 0 & -1\end{pmatrix}$ belgilash kiritib A matritsaning kvadratini hisoblaymiz: $A^2=\begin{pmatrix}1 & 1\\ 0 & -1\end{pmatrix}\cdot\begin{pmatrix}1 & 1\\ 0 & -1\end{pmatrix}=\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$. $A^2=E-birlik\ matritsa.$

Shuning uchun $A^9=\left(A^2\right)^4\cdot A=E^4A=A$ va $\begin{pmatrix}1 & 1\\ 0 & -1\end{pmatrix}^9=\begin{pmatrix}1 & 1\\ 0 & -1\end{pmatrix}$. ▶

? Matritsalarni hisoblang.

1.23. $\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}^5$.

1.24. $\begin{pmatrix}3 & 2\\ -4 & -2\end{pmatrix}^5$.

1.25. $\begin{pmatrix}1 & -2\\ 3 & -4\end{pmatrix}^3$.

1.26. Agar $f(x)=-2x^2+5x+9$, $A=\begin{pmatrix}1 & 2\\ 3 & 0\end{pmatrix}$ bo‘lsa $f(A)$ matritsali ko`phadning qiymatini toping.

Yechish.

$$A^2=A\cdot A=\begin{pmatrix}1 & 2\\ 3 & 0\end{pmatrix}\cdot\begin{pmatrix}1 & 2\\ 3 & 0\end{pmatrix}=\begin{pmatrix}1\cdot1+2\cdot3 & 1\cdot2+2\cdot0\\ 3\cdot1+0\cdot3 & 3\cdot2+0\cdot0\end{pmatrix}=\begin{pmatrix}7 & 2\\ 3 & 6\end{pmatrix},$$

$$f(A)=-2A^2+5A+9E=-2\cdot\begin{pmatrix}7&2\\3&6\end{pmatrix}+5\cdot\begin{pmatrix}1&2\\3&0\end{pmatrix}+9\cdot\begin{pmatrix}1&0\\0&1\end{pmatrix}=$$

$$=\begin{pmatrix}-14&-4\\-6&-12\end{pmatrix}+\begin{pmatrix}5&10\\15&0\end{pmatrix}+\begin{pmatrix}9&0\\0&9\end{pmatrix}=\begin{pmatrix}0&6\\9&-3\end{pmatrix}.$$ ▶

? $f(A)$ matritsali ko`phadning qiymatini toping.

1.27. $f(x)=3x^3+x^2+2,\ A=\begin{pmatrix}1&5\\0&-3\end{pmatrix}.$

1.28. $f(x)=2x^3-3x^2+5,\ A=\begin{pmatrix}1&2\\-2&3\end{pmatrix}.$

1.29. $f(x)=3x^2-5x+2,\ A=\begin{pmatrix}1&2&0\\0&2&-1\\-2&1&4\end{pmatrix}.$

1.30. $f(x)=x^3-6x^2+9x+4,\ A=\begin{pmatrix}1&0&0\\0&2&-1\\0&1&4\end{pmatrix}.$

1.31. $f(x)=x^2+x+1,\ A=\begin{pmatrix}2&1&1\\1&2&1\\1&1&2\end{pmatrix}.$

$A=\begin{pmatrix}1&2&3\\4&5&6\end{pmatrix}$ matritsani transponirlang.

Yechish. $A^T=\begin{pmatrix}1&4\\2&5\\3&6\end{pmatrix}.$ ▶

? Quyidagi matritsalarni transponirlang:

1.32. $A=\begin{pmatrix}3 & 0\\ 2 & -5\end{pmatrix}.$

1.33. $A=\begin{pmatrix}1 & 0\\ -3 & 2\\ 5 & -1\end{pmatrix}.$

1.34. Satrlari ustida elementar almashtirishlar yordamida A matritsani pog‘onasimon ko‘rinishga keltiring:

$$A=\begin{pmatrix}0 & -1 & -1 & -3\\ 1 & 2 & 4 & 7\\ 5 & 0 & 10 & 5\end{pmatrix}$$

Yechish.

$$A=\begin{pmatrix}0 & -1 & -1 & -3\\ 1 & 2 & 4 & 7\\ 5 & 0 & 10 & 5\end{pmatrix} I\leftrightarrow II \sim \begin{pmatrix}1 & 2 & 4 & 7\\ 0 & -1 & -1 & -3\\ 5 & 0 & 10 & 5\end{pmatrix} III-5\cdot I\sim$$

$$\sim\begin{pmatrix}1 & 2 & 4 & 7\\ 0 & -1 & -1 & -3\\ 0 & -10 & -10 & -30\end{pmatrix} III-10\cdot II\sim$$

$$\sim B=\begin{pmatrix}1 & 2 & 4 & 7\\ 0 & -1 & -1 & -3\\ 0 & 0 & 0 & 0\end{pmatrix}-\textit{pog'onasimon matritsa.}$$ ▶

? Matritsalarni pog‘onasimon ko‘rinishga keltiring.

1.35. $\begin{pmatrix}2 & 3 & -2\\ 3 & 1 & 1\\ 1 & 5 & -5\end{pmatrix}.$

1.36. $\begin{pmatrix} 2 & 3 & -2 & 3 \\ 3 & 1 & 1 & 2 \\ 1 & 5 & -5 & 4 \end{pmatrix}$.

1.37. $\begin{pmatrix} 1 & -3 & 1 & 13 \\ 3 & 1 & -7 & 9 \\ -1 & 2 & 0 & -10 \\ 2 & 1 & -5 & 5 \end{pmatrix}$.

1.38. $\begin{pmatrix} 1 & -2 & 1 & 11 \\ 3 & -1 & 2 & 5 \\ 2 & 1 & -3 & -18 \\ 5 & 0 & -1 & -13 \end{pmatrix}$.

1.39. Kopxona 3 xil mahsulot ishlab chiqarish uchun 2 xil xom ashyodan foydalanadi. Xom ashyo xarajatlari $A = \begin{pmatrix} 2 & 3 \\ 5 & 2 \\ 1 & 4 \end{pmatrix}$ matritsa bilan berilgan. Maxsulot ishlab chiqarish rejasi $C = (100 \quad 80 \quad 130)$ - satr-matritsa ko`rinishida berilgan. Har bir xom ashyo turining bir birligi bahosi (pul.birl.) $B = \begin{pmatrix} 30 \\ 50 \end{pmatrix}$ - ustun-matritsa ko`rinishida berilgan. Rejani bajarish uchun sarflanadigan xom ashyo miqdorini va xom ashyoning umumiy bahosini aniqlang.

Yechish. 1-usul. Har bir xom ashyo sarfi

$$S = C \cdot A = (100 \quad 80 \quad 130) \cdot \begin{pmatrix} 2 & 3 \\ 5 & 2 \\ 1 & 4 \end{pmatrix} = (730 \quad 980)$$

bo`lsa, xom ashyoning umumiy bahosi

$$Q = S \cdot B = (C \cdot A) \cdot B = (730 \quad 980) \cdot \begin{pmatrix} 30 \\ 50 \end{pmatrix} = (70900)$$

bo`ladi.

2-usul. Avval har bir mahsulot turiga sarflanuvchi xom ashyo miqdori

$$R = A \cdot B = \begin{pmatrix} 2 & 3 \\ 5 & 2 \\ 1 & 4 \end{pmatrix} \cdot \begin{pmatrix} 30 \\ 50 \end{pmatrix} = \begin{pmatrix} 210 \\ 250 \\ 230 \end{pmatrix}$$

So`ngra, xom ashyoning umumiy bahosini aniqlaymiz

$$Q = C \cdot R = (100 \quad 80 \quad 130) \cdot \begin{pmatrix} 210 \\ 250 \\ 230 \end{pmatrix} = (70900)$$

▶

? Quyidagi iqtisodiy mazmundagi masalalarni yeching:

1.40. Kopxona 3 xil mahsulot ishlab chiqarish uchun 2 xil xom ashyodan foydalanadi. Xom ashyo xarajatlari $A = \begin{pmatrix} 2 & 1 & 3 \\ 1 & 3 & 4 \end{pmatrix}$ matritsa bilan berilgan. Maxsulot ishlab chiqarish rejasi $C = (100 \quad 200 \quad 150)$ – satr-matritsa ko‘rinishida berilgan. Har bir xom ashyo turining bir birligi bahosi (pul.birl.) $B = (10 \quad 15)$ – ustun-matritsa ko‘rinishida berilgan. Rejani bajarish uchun sarflanadigan xom ashyo miqdorini va xom ashyoning umumiy bahosini aniqlang.

1.41. Kopxona 3 xil mahsulot ishlab chiqarish uchun 2 xil xom ashyodan foydalanadi. Xom ashyo xarajatlari $A = \begin{pmatrix} 3 & 1 \\ 2 & 2 \\ 5 & 4 \end{pmatrix}$ matritsa bilan berilgan. Maxsulot ishlab chiqarish rejasi $C = (150 \quad 120 \quad 100)$ – satr-matritsa ko‘rinishida berilgan. Har bir xom ashyo turining bir birligi bahosi (pul.birl.) $B = \begin{pmatrix} 40 \\ 60 \end{pmatrix}$ – ustun-matritsa ko‘rinishida berilgan. Rejani bajarish uchun sarflanadigan xom ashyo miqdorini va xom ashyoning umumiy bahosini aniqlang.

2 DETERMINANTLAR NAZARIYASI

2.1. $\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$ ikkinchi tartibli determinantni hisoblang:

Yechish. $\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 1 \cdot 4 - 2 \cdot 3 = -2.$ ▶

? Ikkinchi tartibli determinantni hisoblang.

2.2. $\begin{vmatrix} -7 & 6 \\ 5 & -4 \end{vmatrix}.$

2.3. $\begin{vmatrix} 10 & -5 \\ 9 & -8 \end{vmatrix}.$

2.4. $\begin{vmatrix} \sqrt{a}+\sqrt{b} & \sqrt{a}-\sqrt{b} \\ \sqrt{a}-\sqrt{b} & \sqrt{a}+\sqrt{b} \end{vmatrix}.$

2.5. $\begin{vmatrix} \sin 1^0 & \sin 89^0 \\ -\cos 1^0 & \cos 89^0 \end{vmatrix}.$

2.6. $\begin{vmatrix} \dfrac{x+y}{x} & \dfrac{2x}{x-y} \\ \dfrac{y-x}{x^2-y^2} & \dfrac{y-x}{x^2-y^2} \end{vmatrix}.$

2.7. $\begin{vmatrix} \sin^2\alpha & \cos^2\alpha \\ \sin^2\beta & \cos^2\beta \end{vmatrix}.$

2.8. $\begin{vmatrix} \sqrt{5}-a^{\frac{1}{2}} & a^{\frac{1}{2}} \\ -a^{\frac{1}{2}} & \sqrt{5}+a^{\frac{1}{2}} \end{vmatrix}.$

2.9. $\begin{vmatrix} \sin 60^0 & \cos 45^0 \\ \sin 45^0 & tg30^0 \end{vmatrix}.$

2.10. $\begin{vmatrix} tga & -1 \\ 4 & ctga \end{vmatrix}.$

2.11. $\begin{vmatrix} 1,(3) & 2,25 \\ 23/3 & 6 \end{vmatrix}.$

2.12. $\begin{vmatrix} \dfrac{a-1}{2\sqrt{a}} & \dfrac{a+\sqrt{a}}{\sqrt{a}-1} \\ \dfrac{a\sqrt{a}-\sqrt{a}}{2a} & \dfrac{a-\sqrt{a}}{\sqrt{a}+1} \end{vmatrix}.$

? Tenglamani yeching.

2.13. $\begin{vmatrix} 2x+1 & 3 \\ x+5 & 2 \end{vmatrix} = 0.$

2.14. $\begin{vmatrix} x+3 & x-1 \\ 7-x & x-1 \end{vmatrix} = 0.$

2.15. $\begin{vmatrix} 2x-1 & x+1 \\ x+2 & x-1 \end{vmatrix} = -6.$

2.16. $\begin{vmatrix} x-2 & y+3 \\ -y-3 & x-2 \end{vmatrix} = 0.$

2.17. $\begin{vmatrix} \sin 2x & \sin x \\ \cos x & \cos 2x \end{vmatrix} = 0.$

2.1. Uchinchi tartibli determinantni hisoblang: $\begin{vmatrix} 3 & 2 & 1 \\ 2 & 5 & 3 \\ 3 & 4 & 2 \end{vmatrix}$.

Yechish. Determinantni birinchi satr elementlari bo‘yicha yoyib hisoblaymiz: $\begin{vmatrix} 3 & 2 & 1 \\ 2 & 5 & 3 \\ 3 & 4 & 2 \end{vmatrix} = 3 \cdot \begin{vmatrix} 5 & 3 \\ 4 & 2 \end{vmatrix} - 2 \cdot \begin{vmatrix} 2 & 3 \\ 3 & 2 \end{vmatrix} + 1 \cdot \begin{vmatrix} 2 & 5 \\ 3 & 4 \end{vmatrix} =$

$$= 3 \cdot (5 \cdot 2 - 3 \cdot 4) - 2 \cdot (2 \cdot 2 - 3 \cdot 3) + 1 \cdot (2 \cdot 4 - 5 \cdot 3) =$$

$$= 3 \cdot (-2) - 2 \cdot (-5) + 1 \cdot (-7) = -3.$$

Uchinchi tartibli determinantlarni ixtiyoriy satr (ustun) elementlari bo‘yicha yoyib hisoblang.

2.2. $\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$.

2.3. $\begin{vmatrix} 2 & 1 & 3 \\ 5 & 3 & 2 \\ 1 & 4 & 3 \end{vmatrix}$.

2.4. $\begin{vmatrix} 3 & 2 & -1 \\ -2 & 2 & 3 \\ 4 & 2 & -3 \end{vmatrix}$.

2.5. $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 6 \end{vmatrix}$.

2.6. $\begin{vmatrix} 2 & 1 & 3 \\ 4 & 5 & 9 \\ 16 & 25 & 81 \end{vmatrix}$.

2.7. Uchinchi tartibli determinantni uchburchak qoidasidan foydalanib hisoblang: $\begin{vmatrix} 1 & 2 & 3 \\ -4 & 5 & -6 \\ 7 & 8 & 9 \end{vmatrix}$.

Yechish. $\begin{vmatrix} 1 & 2 & 3 \\ -4 & 5 & -6 \\ 7 & 8 & 9 \end{vmatrix} = 1\cdot 5\cdot 9 + 2\cdot(-6)\cdot 7 + (-4)\cdot 3\cdot 8 -$

$-3\cdot 5\cdot 7 - (-4)\cdot 2\cdot 9 - 1\cdot(-6)\cdot 7 = 45 - 84 - 96 - 105 + 72 + 42 = 126.$ ▶

? Uchburchak qoidasidan foydalanib determinantlarni hisoblang.

2.8. $\begin{vmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 0 \end{vmatrix}$.

2.9. $\begin{vmatrix} 0 & x & 0 \\ y & 0 & 0 \\ 0 & 0 & z \end{vmatrix}$.

2.10. $\begin{vmatrix} 0 & 1 & 0 \\ 2 & 3 & 4 \\ 0 & 5 & 0 \end{vmatrix}$.

? Uchinchi tartibli determinantlarni qulay usulda hisoblang.

2.11. $\begin{vmatrix} 2 & 3 & 4 \\ 5 & -2 & 1 \\ 1 & 2 & 3 \end{vmatrix}$.

2.12. $\begin{vmatrix} a & 1 & a \\ -1 & a & 1 \\ a & -1 & a \end{vmatrix}$.

2.13. $\begin{vmatrix} 1 & 2 & 3 \\ 8 & 1 & 4 \\ 2 & 1 & 1 \end{vmatrix}$.

2.14. $\begin{vmatrix} 5 & 3 & 2 \\ -1 & 2 & 4 \\ 7 & 3 & 6 \end{vmatrix}$.

2.15. $\begin{vmatrix} 3 & -1 & -2 \\ 1 & 2 & 5 \\ -4 & 1 & 6 \end{vmatrix}$.

2.16. $\begin{vmatrix} 1 & 2 & -1 \\ 3 & 7 & 2 \\ 2 & 3 & -7 \end{vmatrix}$.

2.17. $\begin{vmatrix} a & -a & a \\ a & a & -a \\ a & -a & -a \end{vmatrix}$.

2.18. $\begin{vmatrix} a+x & x & x \\ x & b+x & x \\ x & x & c+x \end{vmatrix}$.

2.19. $\begin{vmatrix} \cos\alpha & \sin\alpha\cos\beta & \sin\alpha\sin\beta \\ -\sin\alpha & \cos\alpha\cos\beta & \cos\alpha\sin\beta \\ 0 & -\sin\beta & \cos\beta \end{vmatrix}$.

2.20. $\begin{vmatrix} x^2 & x & 1 \\ y^2 & y & 1 \\ z^2 & z & 1 \end{vmatrix}.$

2.21. $\begin{vmatrix} m+a & m-a & a \\ n+a & 2n-a & a \\ a & -a & a \end{vmatrix}.$

2.22. $\begin{vmatrix} ax & a^2+x^2 & 1 \\ ay & a^2+y^2 & 1 \\ az & a^2+z^2 & 1 \end{vmatrix}.$

2.23. $\begin{vmatrix} \sin 3\alpha & \cos 3\alpha & 1 \\ \sin 2\alpha & \cos 2\alpha & 1 \\ \sin\alpha & \cos\alpha & 1 \end{vmatrix}.$

2.24. $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}.$

2.25. $\begin{vmatrix} a & x & x \\ x & b & x \\ x & x & c \end{vmatrix}.$

? Tenglama va tengsizlikni yeching.

2.26. $\begin{vmatrix} 2 & 0 & 3 \\ -1 & 7 & x-3 \\ 5 & -3 & 6 \end{vmatrix} = 0.$

2.27. $\begin{vmatrix} -1 & 3 & -2 \\ 2-3x & 0 & 5 \\ 3 & 2 & 1 \end{vmatrix} \geq 0.$

2.28. $\begin{vmatrix} -1 & 0 & 2x+3 \\ 3-x & 1 & 1 \\ 2x+1 & -1 & 2 \end{vmatrix} = 0.$

2.29. $\begin{vmatrix} 6 & 3 & x-1 \\ 2x & 1 & 0 \\ 4 & x+2 & 2 \end{vmatrix} = 0.$

2.30. Determinantning xossalaridan foydalanib tenglikni isbotlang:

$$\begin{vmatrix} a_1 & b_1 & c_1+a_1x+b_1y \\ a_2 & b_2 & c_2+a_2x+b_2y \\ a_3 & b_3 & c_3+a_3x+b_3y \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}.$$

Isbot. Chap determinantning uchunchi ustunini uchta ustun yig‘indisi ko‘rinishida ifodalash mumkin, bu determinantni uchta determinant yig‘indisi ko‘rinishida ifodalaymiz:

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} a_1 & b_1 & a_1x \\ a_2 & b_2 & a_2x \\ a_3 & b_3 & a_3x \end{vmatrix} + \begin{vmatrix} a_1 & b_1 & b_1y \\ a_2 & b_2 & b_2y \\ a_3 & b_3 & b_3y \end{vmatrix}.$$

Ikkinchi determinantning uchunchi ustuni birinchi ustuniga proporsional, uchunchi determinantning uchunchi ustuni ikkinchi ustuniga proporsional. Shuning uchun ikkinchi va uchinchi determinantlar nolga teng. ▶

?

Tenglikni isbotlang.

2.31. $\begin{vmatrix} a_1+b_1x & a_1-b_1x & c_1 \\ a_2+b_2x & a_2-b_2x & c_2 \\ a_3+b_3x & a_3-b_3x & c_3 \end{vmatrix} = -2x \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}.$

2.32. $\begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} = (b-a)(c-a)(c-b).$

2.33. $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (b-a)(c-a)(c-b).$

2.34. $\begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} = \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}.$

2.35. $\begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = (a+b+c)\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}.$

2.36. $\begin{vmatrix} (a+b)^2 & c^2 & c^2 \\ a^2 & (b+c)^2 & a^2 \\ b^2 & b^2 & (c+a)^2 \end{vmatrix} = 2bc(a+b+c)^3.$

2.37. To‘rtinchi tartibli determinantni hisoblang:

$$\Delta = \begin{vmatrix} a & 0 & 3 & 5 \\ 0 & 0 & b & 2 \\ 1 & c & 2 & 3 \\ 0 & 0 & 0 & d \end{vmatrix}.$$

Determinantni to‘rtinchi satr elementlari bo‘yicha yoyib hisoblaymiz:

Yechish. $\Delta = (+d)\cdot\begin{vmatrix} a & 0 & 3 \\ 0 & 0 & b \\ 1 & c & 2 \end{vmatrix} = \begin{bmatrix} \text{det}\,er\,\text{min}\,antni \\ 2-satr\,bo'yicha\,yoyamiz \end{bmatrix} =$

$= d\cdot(-b)\cdot\begin{vmatrix} a & 0 \\ 1 & c \end{vmatrix} = -d\cdot b\cdot a\cdot c.$ ▶

?

To‘rtinchi tartibli determinantni hisoblang.

2.38. $\begin{vmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & 2 & 3 \\ 3 & 0 & 1 & 2 \\ 2 & 3 & 0 & 1 \end{vmatrix}.$

2.39. $\begin{vmatrix} 1 & 2 & 0 & -3 \\ 3 & 1 & 0 & 4 \\ 1 & 5 & -1 & 7 \\ -2 & 1 & 0 & 1 \end{vmatrix}.$

2.40. $\begin{vmatrix} 1 & 2 & 3 & 4 \\ -9 & -9 & -9 & -9 \\ 4 & 3 & 2 & 1 \\ 1 & 0 & 1 & 0 \end{vmatrix}.$

2.41. $\begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & 2 & 5 & 9 \\ 0 & 0 & 3 & 7 \\ -2 & -4 & -6 & 1 \end{vmatrix}.$

2.42. $\begin{vmatrix} 1 & 2 & -3 & 1 \\ 3 & 0 & 1 & -1 \\ 2 & 0 & 4 & 1 \\ 5 & 1 & 2 & 1 \end{vmatrix}.$

2.43. $\begin{vmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{vmatrix}.$

2.44. $\begin{vmatrix} 1 & -2 & 3 & 4 \\ 2 & 1 & -4 & 3 \\ 3 & -4 & -1 & -2 \\ 4 & 3 & 2 & -1 \end{vmatrix}.$

2.45. $\begin{vmatrix} -1 & -1 & -1 & -1 \\ -1 & -2 & -4 & -8 \\ -1 & -3 & -9 & -27 \\ -1 & -4 & -16 & -64 \end{vmatrix}.$

2.46. $\begin{vmatrix} 1+a & 1 & 1 & 1 \\ 1 & 1-a & 1 & 1 \\ 1 & 1 & 1+b & 1 \\ 1 & 1 & 1 & 1-b \end{vmatrix}.$

2.47. $\begin{vmatrix} 3 & -1 & 2 & -1 & 1 \\ 5 & 1 & -2 & 1 & 2 \\ 9 & -1 & 1 & 3 & 4 \\ 3 & 0 & 6 & -1 & 3 \\ 5 & 2 & 3 & -2 & 1 \end{vmatrix}.$

2.48. $\begin{vmatrix} 0 & 6 & 3 & 5 & 1 \\ -3 & 2 & 4 & 1 & 0 \\ 5 & 1 & 4 & 3 & 2 \\ -3 & 8 & 7 & 6 & 1 \\ 1 & 0 & 3 & 4 & 0 \end{vmatrix}.$

3 MATRITSA RANGI. TESKARI MATRITSA

3.1. $A=\begin{pmatrix}2 & 3 & 2\\ 5 & 1 & 4\\ 1 & -2 & -1\end{pmatrix}$ matritsa uchun teskari A^{-1} matritsani klassik usulda toping.

Yechish. Klassik usulda teskari matritsa

$$A^{-1}=\frac{1}{|A|}\begin{pmatrix}A_{11} & A_{21} & A_{31}\\ A_{12} & A_{22} & A_{32}\\ A_{13} & A_{23} & A_{33}\end{pmatrix} \qquad (1)$$

formula bo‘yicha hisoblanadi.

Bu yerda $|A|$ berilgan matritsa determinanti,

$A_{ij}\ (i=1,2,3;\ j=1,2,3)$ A matritsa elementlarining algebraik to‘ldiruvchilari.

$$|A|=\begin{vmatrix}2 & 3 & 2\\ 5 & 1 & 4\\ 1 & -2 & -1\end{vmatrix}=-2+12-20-2+15+16=43-24=19\neq 0$$

Demak A xosmas matritsa, va A^{-1} teskari matritsa mavjud.

Algebraik to‘ldiruvchilarni hisoblaymiz:

$$A_{11}=\begin{vmatrix}1 & 4\\ -2 & -1\end{vmatrix}=-1+8=7;\quad A_{12}=-\begin{vmatrix}5 & 4\\ 1 & -1\end{vmatrix}=-(-5-4)=9;$$

$$A_{13}=\begin{vmatrix}5 & 1\\ 1 & -2\end{vmatrix}=-10-1=-11;\quad A_{21}=-\begin{vmatrix}3 & 2\\ -2 & -1\end{vmatrix}=-(-3+4)=-1;$$

$$A_{22}=\begin{vmatrix}2 & 2\\ 1 & -1\end{vmatrix}=-2-2=-4;\quad A_{23}=-\begin{vmatrix}2 & 3\\ 1 & -2\end{vmatrix}=-(-4-3)=7;$$

$$A_{31}=\begin{vmatrix}3 & 2\\ 1 & 4\end{vmatrix}=12-2=10;\quad A_{32}=-\begin{vmatrix}2 & 2\\ 5 & 4\end{vmatrix}=-(8-10)=2;$$

$$A_{33}=\begin{vmatrix}2 & 3\\ 5 & 1\end{vmatrix}=2-15=-13;$$

A_{ij} larni (1) formulaga qo‘yamiz va teskari

$$A^{-1}=1/19\begin{pmatrix}7 & -1 & 10\\ 9 & -4 & 2\\ -11 & 7 & -13\end{pmatrix}$$ matritsani olamiz.

Teskari matritsaning to’griligini tekshirish uchun quyidagi tenglikni tekshiramiz:

$$AA^{-1}=A^{-1}A=E \qquad (2)$$

$$\begin{pmatrix}2 & 3 & 2\\ 5 & 1 & 4\\ 1 & -2 & -1\end{pmatrix}\cdot 1/19\begin{pmatrix}7 & -1 & 10\\ 9 & -4 & 2\\ -11 & 7 & -13\end{pmatrix}=$$

$$=1/19\cdot\begin{pmatrix}14+27-22 & -2-12+14 & 20+6-26\\ 35+9-44 & -5-4+28 & 50+2-52\\ 7-18+11 & -1+8-7 & 10-4+13\end{pmatrix}=$$

$$=1/19\cdot\begin{pmatrix}19 & 0 & 0\\ 0 & 19 & 0\\ 0 & 0 & 19\end{pmatrix}=\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}=F$$

Demak, A^{-1} to‘g‘ri topilgan. ▶

3.2. $A=\begin{pmatrix}1 & 2 & 1\\ -1 & -1 & -3\\ 4 & 3 & -2\end{pmatrix}$ matritsa uchun A^{-1} matritsani Gauss-Jordan usulida toping.

Yechish. $|A|=16\neq 0$ teskari matritsa mavjud. Berilgan matritsani birlik matritsa hisobida kengaytirib, elementar almashtirishlar bajaramiz, bu usulni to chap tomonda A matritsa o‘rnida birlik matritsa hosil bo‘lguncha davom ettiramiz, o‘ng tomonda hosil bo‘lgan matritsa berilgan matritsaga nisbatan teskari matritsa bo‘ladi.

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ -1 & -1 & -3 & 0 & 1 & 0 \\ 4 & 3 & -2 & 0 & 0 & 1 \end{array}\right) \sim \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -2 & 1 & 1 & 0 \\ 0 & -5 & -6 & -4 & 0 & 1 \end{array}\right) \sim$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -2 & 1 & 1 & 0 \\ 0 & 0 & -16 & 1 & 5 & 1 \end{array}\right) \sim \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -2 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1/16 & -5/16 & -1/16 \end{array}\right) \sim$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 5 & -1 & -2 & 0 \\ 0 & 1 & 0 & 14/16 & 6/16 & -2/16 \\ 0 & 0 & 1 & -1/16 & -5/16 & -1/16 \end{array}\right) \sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -11/16 & -7/16 & 5/16 \\ 0 & 1 & 0 & 14/16 & 6/16 & -2/16 \\ 0 & 0 & 1 & -1/16 & -5/16 & -1/16 \end{array}\right)$$

$A^{-1} = 1/16 \begin{pmatrix} -11 & -7 & 5 \\ 14 & 6 & -2 \\ -1 & -5 & -1 \end{pmatrix}$ teskari matritsa to‘g‘ri topilganini (2) formulaga qo‘yib

tekshiramiz: $AA^{-1} = 1/16 \begin{pmatrix} 1 & 2 & 1 \\ -1 & -1 & -3 \\ 4 & 3 & -2 \end{pmatrix} \cdot \begin{pmatrix} -11 & -7 & -5 \\ 14 & 6 & -2 \\ -1 & -5 & -1 \end{pmatrix} =$

$$= 1/16 \begin{pmatrix} -11+28-1 & -7+12-5 & 5-4-1 \\ 11-14+3 & 7-6+15 & -5+2+3 \\ -44+42+2 & -28+18+10 & 20-6+2 \end{pmatrix} =$$

$= 1/16 \begin{pmatrix} 16 & 0 & 0 \\ 0 & 16 & 0 \\ 0 & 0 & 16 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ demak, teskari matritsa to‘gri topilgan.

▶

? Teskari matritsani toping.

3.3. Berilgan kvadrat matritsalar uchun teskari matritsani ikki usulda toping:

$$a)\begin{pmatrix} -1 & 1 \\ 4 & 2 \end{pmatrix}; \quad b)\begin{pmatrix} 1 & 3 \\ 2 & 6 \end{pmatrix}; \quad c)\begin{pmatrix} tg\alpha & 1 \\ 2 & ctg\alpha \end{pmatrix}; \quad d)\begin{pmatrix} 2 & 1 & 1 \\ 1 & 0 & 2 \\ 3 & 1 & 2 \end{pmatrix}$$

$$e)\begin{pmatrix} 1 & -1 & 1 \\ -38 & 41 & -34 \\ 27 & -29 & 24 \end{pmatrix};\quad f)\begin{pmatrix} 2 & 5 & 7 \\ 6 & 3 & 4 \\ 5 & -2 & -3 \end{pmatrix};\quad j)\begin{pmatrix} 3 & -4 & 5 \\ 2 & -3 & 1 \\ 3 & -5 & -1 \end{pmatrix}$$

3.4. Berilgan kvadrat matritsalar uchun teskari matritsani qulay usulda toping:

$$a)\begin{pmatrix} 1 & 5 & 7 \\ 3 & 1 & 1 \\ 2 & 3 & 4 \end{pmatrix};\quad b)\begin{pmatrix} 2 & -1 & 7 \\ 5 & 3 & 2 \\ 1 & 4 & 3 \end{pmatrix};\quad c)\begin{pmatrix} 1 & 0 & -2 \\ 3 & 1 & 0 \\ -1 & 2 & 4 \end{pmatrix};\quad e)\begin{pmatrix} 3 & 2 & 2 \\ 1 & 3 & 1 \\ 5 & 3 & 4 \end{pmatrix}$$

$$d)\begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ 0 & 1 & 1 & \dots & 1 \\ 0 & 0 & 1 & \dots & 1 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix};\quad f)\begin{pmatrix} 1 & 0 & 0 & 0 & \dots & 0 & 0 \\ a & 1 & 0 & 0 & \dots & 0 & 0 \\ 0 & a & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & a & 1 & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & 0 & \dots & a & 1 \end{pmatrix};$$

$$g)\begin{pmatrix} 1 & a & a^2 & a^3 & \dots & a^n \\ 0 & 1 & a & a^2 & \dots & a^{n-1} \\ 0 & 0 & 1 & a & \dots & a^{n-2} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & 0 & \dots & 1 \end{pmatrix}$$

3.5. Matritsa rangini ta'rifga asosan hisoblang: $A=\begin{pmatrix} 2 & -1 & 3 & -2 & 4 \\ 4 & -2 & 5 & 1 & 7 \\ 2 & -1 & 1 & 8 & 2 \end{pmatrix}$.

Yechish. A matritsa $3-$tartibli, demak uning rangi 3 dan yuqori bo'lmaydi. Uchinchi tartibli minorlarni hisoblaymiz:

$$M_1=\begin{vmatrix} 2 & -1 & 3 \\ 4 & -2 & 5 \\ 2 & -1 & 1 \end{vmatrix}=-4-10-12+12+4+10=0$$

$$M_2=\begin{vmatrix} 2 & -1 & -2 \\ 4 & -2 & 1 \\ 2 & -1 & 8 \end{vmatrix}=-32-2+8-8+32+2=0$$

$$M_3=\begin{vmatrix} 2 & -1 & 4 \\ 4 & -2 & 7 \\ 2 & -1 & 2 \end{vmatrix}=-8-14-16+16+8-14=0$$

$$M_4 = \begin{vmatrix} -1 & 3 & -2 \\ -2 & 5 & 1 \\ -1 & 1 & 8 \end{vmatrix} = -40 - 3 + 4 - 10 + 48 + 1 = 0$$

$$M_5 = \begin{vmatrix} 3 & -2 & 4 \\ 5 & 1 & 7 \\ 1 & 8 & 2 \end{vmatrix} = 6 - 14 + 160 - 4 + 20 - 168 = 0$$

Barcha uchinchi tartibli minorlar nolga teng. Ikkinchi tartibli minorlarni hisoblaymiz:

$$M_1^1 = \begin{vmatrix} -1 & 3 \\ -2 & 5 \end{vmatrix} = -5 + 6 = 1 \qquad M_1^1 \neq 0, \qquad r(A) = 2.$$

Bu usulda noldan farqli minor topilgunga qadar hisoblashlar davom etadi. Shuning uchun $3-$tartibli kattaroq matritsa rangini hisoblash bir muncha qiyinchiliklarga olib keladi. ▶

3.6. Matritsa rangini elementar almashtirishlar yordamida nollar yig‘ib hisoblang:

$$A = \begin{pmatrix} 25 & 31 & 17 & 43 \\ 75 & 94 & 53 & 132 \\ 75 & 94 & 54 & 134 \\ 25 & 32 & 20 & 48 \end{pmatrix}$$

Yechish. $$A = \begin{pmatrix} 25 & 31 & 17 & 43 \\ 75 & 94 & 53 & 132 \\ 75 & 94 & 54 & 134 \\ 25 & 32 & 20 & 48 \end{pmatrix} \sim \begin{pmatrix} 25 & 31 & 17 & 43 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 3 & 5 \\ 0 & 1 & 3 & 5 \end{pmatrix} \sim \begin{pmatrix} 25 & 31 & 17 & 43 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Bu matritsaning rangi $\begin{pmatrix} 25 & 31 & 17 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$ matritsa rangiga teng.

$$\begin{vmatrix} 25 & 31 & 17 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{vmatrix} = 25 \neq 0 \qquad r\begin{pmatrix} 25 & 31 & 17 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} = 3$$

Demak, berilgan matritsaning rangi ham 3 ga teng. $r(A)=3$. ▶

? Mashqlarni bajaring.

3.7. Berilgan kvadrat matritsaning rangini toping. Xosmas matritsaning teskarisini toping:

$$a)\ A=\begin{pmatrix}1 & 2\\ -2 & 0\end{pmatrix};\qquad b)\ B=\begin{pmatrix}2 & 5\\ -4 & 2\end{pmatrix};$$

$$c)\ C=\begin{pmatrix}-1 & 0 & 8\\ 5 & 9 & 0\\ 0 & 4 & 3\end{pmatrix};\qquad d)\ D=\begin{pmatrix}2 & 1 & 2\\ 1 & 1 & 1\\ 2 & 3 & 2\end{pmatrix};$$

$$f)\ F=\begin{pmatrix}2 & 3 & 4 & 0\\ 1 & 5 & 7 & 0\\ 3 & 1 & 1 & 0\\ 0 & 0 & 0 & 1\end{pmatrix};\qquad g)\ G=\begin{pmatrix}1 & 2 & 1 & 0\\ 1 & 1 & 3 & 1\\ 1 & 2 & 1 & 1\\ 1 & 1 & 3 & 0\end{pmatrix}$$

3.8. Quyidagi matritsalar rangini minorlar ajratish usuli bilan hisoblang:

$$a)\begin{pmatrix}3 & 5 & 7\\ 1 & 2 & 3\\ 1 & 3 & 5\end{pmatrix};b)\begin{pmatrix}1 & 2 & 3 & 6\\ 2 & 3 & 1 & 6\\ 3 & 1 & 2 & 6\end{pmatrix};c)\begin{pmatrix}1 & 2 & 3 & 4\\ 2 & 4 & 6 & 8\\ 3 & 6 & 9 & 12\end{pmatrix};d)\begin{pmatrix}0 & 2 & 0 & 0\\ 1 & 0 & 0 & 4\\ 0 & 0 & 3 & 0\end{pmatrix}$$

$$e)\begin{pmatrix}2 & -4 & 3 & 1 & 0\\ 1 & -2 & 1 & -4 & 2\\ 0 & 1 & -1 & 3 & 1\\ 4 & -7 & 4 & -4 & 5\end{pmatrix};j)\begin{pmatrix}1 & 2 & 1 & 3\\ 4 & -1 & -5 & -6\\ 1 & -3 & -4 & -7\\ 2 & 1 & -1 & 0\end{pmatrix};k)\begin{pmatrix}0 & 2 & -4\\ -1 & -4 & 5\\ 3 & 1 & 7\\ 0 & 5 & -10\\ 2 & 3 & 0\end{pmatrix}$$

? Quyidagi 3.9 – 3.19 – misollarda matritsalar rangini elementar almashtirish usuli bilan hisoblang:

3.9. $\begin{pmatrix} 1 & 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 6 & 8 \\ 1 & 2 & 1 & 8 & 4 \end{pmatrix}.$

3.10. $\begin{pmatrix} 1 & 7 & 5 & 8 & 9 & 2 \\ 3 & 21 & 15 & 24 & 27 & 6 \\ 2 & 14 & 10 & 16 & 18 & 4 \end{pmatrix}.$

3.11. $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \end{pmatrix}.$

3.12. $\begin{pmatrix} 1 & 0 & 2 & 0 & 0 \\ 0 & 1 & 0 & 2 & 0 \\ 2 & 0 & 4 & 0 & 0 \end{pmatrix}.$

3.13. $\begin{pmatrix} 4 & 3 & -5 & 2 & 3 \\ 8 & 6 & -7 & 4 & 2 \\ 4 & 3 & -8 & 2 & 7 \\ 4 & 3 & 1 & 2 & -5 \\ 8 & 6 & -1 & 4 & -6 \end{pmatrix}.$

3.14. $\begin{pmatrix} 24 & 19 & 36 & 72 & -38 \\ 49 & 40 & 73 & 147 & -80 \\ 73 & 59 & 98 & 219 & -118 \\ 47 & 36 & 71 & 141 & -72 \end{pmatrix}.$

3.15. $\begin{pmatrix} 17 & -28 & 45 & 11 & 39 \\ 24 & -37 & 61 & 13 & 50 \\ 25 & -7 & 32 & -18 & -11 \\ 31 & 12 & 19 & -43 & -55 \\ 42 & 13 & 29 & -55 & -68 \end{pmatrix}.$

3.16. $\begin{pmatrix} 47 & -67 & 35 & 201 & 155 \\ 26 & 98 & 23 & -294 & 6 \\ 16 & -428 & 1 & 1284 & 52 \end{pmatrix}.$